

Lab Assessment Rubrics of Digital System Design (CPE-344)

Lab Number			08							
Lab Title			Implementation of carry look ahead adder through full adders in Verilog HDL							
Submitted by										
Names			Registration Number							
Sumaira Safeer			FA22-BCE-019							
Rubrics name & number								Marks		
								In-Lab	Post-Lab	
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.									
	R3: Interpretation of Subject Knowledge: Ability to interpret and explain mathematical and/or visual forms, including equations, diagrams, graphics, figures, and tables.									
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.									
Design	R7: Implementing Design Strategy: Ability to execute a solution taking into consideration design requirements and pertinent contextual elements. [Block Diagram/Flow chart/Circuit Diagram]									
	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.									
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.									
	R11: Tools Evaluation: Ability to simulate the experiment and them using hardware tools to verify The results.									
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.									
	R13: Management of Teamwork: Ability to appreciate, understand and work with multidisciplinary team members.									
Rubrics to follow										
Rubrics	R2	R3	R5	R6	R7	R8	R9	R11	R12	R13
In-Lab										
Post-Lab										

Lab: 08

IN Lab Tasks:

Task#01

- **Verilog Code**

```
module carryLookAheadAdder4Bit (
    output [3:0] sum,
    output      c_out,
    input  [3:0] a,
    input  [3:0] b,
    input      c_in
);

    // Internal wires
    wire p0, g0, p1, g1, p2, g2, p3, g3;
    wire c1, c2, c3, c4;

    // Generate (G) signals
    assign g0 = a[0] & b[0];
    assign g1 = a[1] & b[1];
    assign g2 = a[2] & b[2];
    assign g3 = a[3] & b[3];

    // Propagate (P) signals
    assign p0 = a[0] ^ b[0];
    assign p1 = a[1] ^ b[1];
    assign p2 = a[2] ^ b[2];
    assign p3 = a[3] ^ b[3];

    // Carry Lookahead logic
    // Note: c_in is equivalent to c0
    assign c1 = g0 | (p0 & c_in);
    assign c2 = g1 | (p1 & g0) | (p1 & p0 & c_in);
    assign c3 = g2 | (p2 & g1) | (p2 & p1 & g0) | (p2 & p1 & p0 & c_in);
    assign c4 = g3 | (p3 & g2) | (p3 & p2 & g1) |
        (p3 & p2 & p1 & g0) | (p3 & p2 & p1 & p0 & c_in);

    // Sum bits
    assign sum[0] = p0 ^ c_in;
    assign sum[1] = p1 ^ c1;
    assign sum[2] = p2 ^ c2;
    assign sum[3] = p3 ^ c3;

    // Final carry out
    assign c_out = c4;
endmodule
```

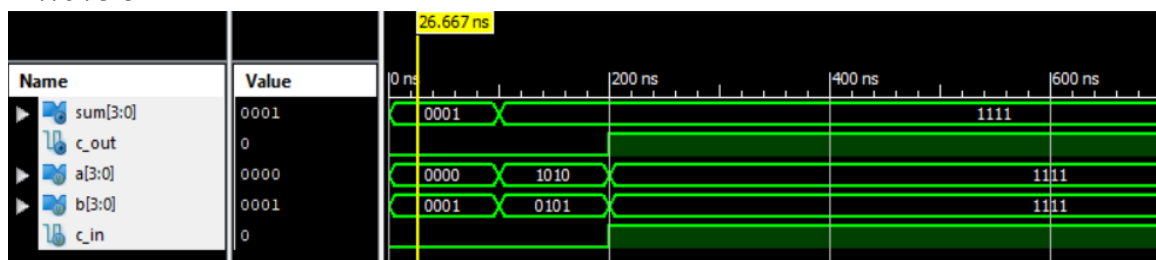
- **Verilog Test Bench**

```

module carryLookAheadAdder4BitTestBench;
    reg [3:0] a;
    reg [3:0] b;
    reg c_in;
    wire [3:0] sum;
    wire c_out;
    carryLookAheadAdder4Bit uut (
        .sum(sum),
        .c_out(c_out),
        .a(a),
        .b(b),
        .c_in(c_in)
    );
    initial begin
        a = 4'b0000;
        b = 4'b0001;
        c_in = 0;
        #100;
        a = 4'b1010;
        b = 4'b0101;
        c_in = 0;
        #100;
        a = 4'b1111;
        b = 4'b1111;
        c_in = 1;
    end
endmodule

```

- **Waveform**



Task#02

- **Verilog Code**

```

module carryLookAheadAdder6Bit(sum, c_out, a, b, c_in);
    //inputs and output
    output [5:0]sum;
    output c_out;

```

```

input [5:0]a, b;
input c_in;

wire p0, g0, p1, g1, p2, g2, p3, g3, p4, g4, p5, g5;
wire c6, c5, c4, c3, c2, c1;

//compute the g for each stage
assign g0 = a[0] & b[0],
g1 = a[1] & b[1],
      g2 = a[2] & b[2],
      g3 = a[3] & b[3],
      g4 = a[4] & b[4],
      g5 = a[5] & b[5];

assign p0 = a[0] ^ b[0],
p1 = a[1] ^ b[1],
p2 = a[2] ^ b[2],
p3 = a[3] ^ b[3],
p4 = a[4] ^ b[4],
p5 = a[5] ^ b[5];
assign c1 = g0 | (p0 & c_in),
c2 = g1 | (c1 & p1),
c3 = g2 | (c2 & p2),
c4 = g3 | (c3 & p3),
c5 = g4 | (c4 & p4),
c6 = g5 | (c5 & p5);

assign sum[0] = p0 ^ c_in,
sum[1] = p1 ^ c1,
sum[2] = p2 ^ c2,
sum[3] = p3 ^ c3,
sum[4] = p4 ^ c4,
sum[5] = p5 ^ c5;

assign c_out = c6;
endmodule

```

- **Verilog Test Bench**

```

module carryLookAheadAdder6BitTestBench;
reg [5:0] a;
reg [5:0] b;
reg c_in;
wire [5:0] sum;
wire c_out;
carryLookAheadAdder6Bit uut (
    .sum(sum),
    .c_out(c_out),
    .a(a),
    .b(b),

```

```

        .c_in(c_in)
    );
    initial begin
        a = 6'B000000;
        b = 6'B010101;
        c_in = 0;
        #100;
        a = 6'B101010;
        b = 6'B010101;
        c_in = 0;
        #100;
        a = 6'B111111;
        b = 6'B000000;
        c_in = 1;
        #100;
        a = 6'B111111;
        b = 6'B111111;
        c_in = 1;
    end
endmodule

```

- **Waveform**

